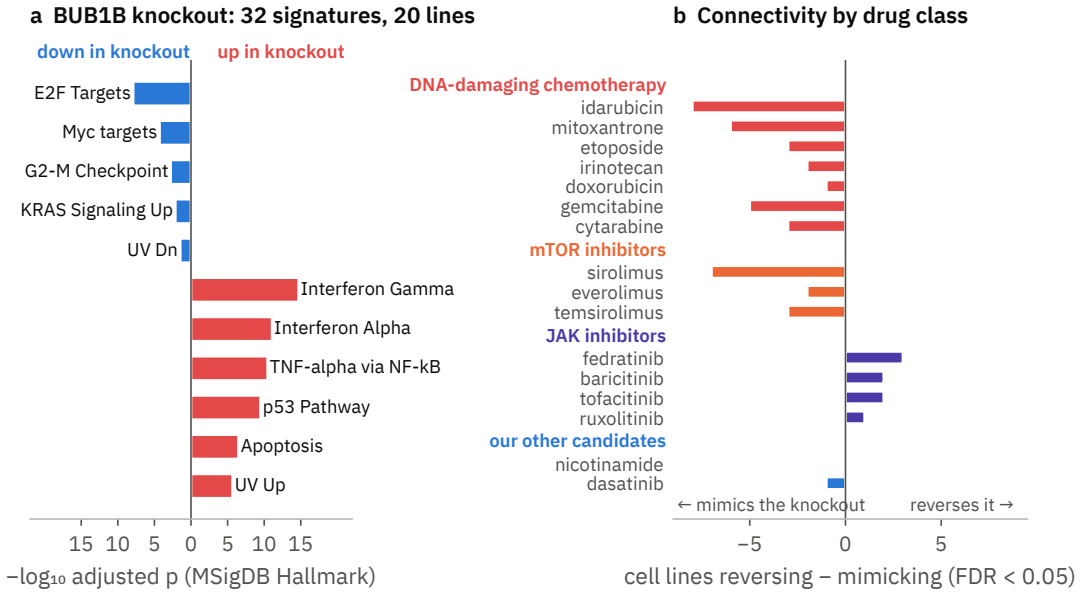
a BUB1B knockout: 32 signatures, 20 lines
down in knockout
up in knockout
E2F Targets
Myc targets
G2-M Checkpoint
KRAS Signaling Up
UV Dn
Interferon Gamma
Interferon Alpha
TNF-alpha via NF-kB
p53 Pathway
Apoptosis
UV Up
15 10 5 0 5 10 15
−log10 adjusted p (MSigDB Hallmark)
b Connectivity by drug class
DNA-damaging chemotherapy
idarubicin
mitoxantrone
etoposide
irinotecan
doxorubicin
gemcitabine
cytarabine
mTOR inhibitors
sirolimus
everolimus
temsirolimus
JAK inhibitors
fedratinib
baricitinib
tofacitinib
ruxolitinib
our other candidates
nicotinamide
dasatinib
← mimics the knockout
reverses it →
−5 0 5
cell lines reversing − mimicking (FDR < 0.05)